

KITABU QUEST

S2

MATHEMATICS



Cover scene

learners solving mathematics with charts and measuring tools with a mountain gorilla mascot

● Clear lessons

● Worked examples

● Fun activities

● Real-life problems

GRADE

2

Title Page

KITABU QUEST S2 Mathematics

Rwanda REB-CBC learner book

Mascot: mountain gorilla

This KITABU QUEST book turns the curriculum into short lessons, worked examples, guided activities, practice, review, and home links for Rwanda learners.

How To Use This Book

This book helps S2 learners study Mathematics one teachable topic at a time.

1. Start with the inquiry question and say what you already know.
2. Read the Learn section slowly and mark new words.
3. Try the worked example before reading the full solution.
4. Complete the activity and practice tasks.
5. Correct your answer using the success criteria.
6. Ask for support when your answer is still unclear.

Subject method: Use objects, drawings, number sentences, and worked examples before independent practice.

Learning Skills In This Book

Each topic builds knowledge, skill, values, communication, collaboration, critical thinking, creativity, digital awareness, and self-confidence.

Study actively: speak answers aloud, draw when useful, show your working, cite evidence, use local examples, and correct mistakes after feedback.

A strong answer is specific. It names the idea, gives evidence or method, applies it to the task, and checks whether it answers the question.

Table Of Contents

Use this table to move through the book one topic at a time. Each topic has a lesson opener, clear teaching, a model or example, guided work, practice, and checks for understanding.

1. Indices and surds
2. Polynomials
3. Simultaneous linear equations and inequalities
4. Multiplier for proportional change
5. Thales theorem
6. Pythagoras's theorem
7. Vectors
8. Parallel and orthogonal projections
9. Isometries
10. Statistics (grouped data)
11. Tree and Venn diagrams

As you finish each topic, return to the success criteria and correct one answer before moving on.

Indices and surds: Lesson Opener

Learning focus: Indices and surds

Country link: a Rwandan school, market, farming, building, transport, or data problem for Indices and surds.

Progression focus: connect ideas across topics and explain causes, methods, or consequences; show methods, units, checks, and a final explanation.

By the end of this topic, you should be able to:

- to be able to determine the probability of an event happening using equally likely events or experiments
- recognize laws of indices
- represent very small numbers or large numbers in standard form
- define and give examples of surds
- identify properties of surds
- recognize the conjugates of surds

Inquiry questions:

- How can indices and surds help you solve a real problem in Mathematics?

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Indices and surds: Learn

Indices and surds teaches how likely an event is.

Core idea:

- Some events are certain, likely, unlikely, or impossible.
- A simple experiment can help you compare chances.
- Record results before making a conclusion.

Key words:

- Indices and surds
- Indices
- surds
- method
- model
- unit
- answer
- check

Indices and surds: Worked Example

Worked example for Indices and surds: A bag has 4 red counters and 1 blue counter. Which colour is more likely to be picked?

1. Compare the number of counters.
2. Red has 4 chances; blue has 1 chance.
3. More red counters means red is more likely.

Answer: Red is more likely.

Indices and surds: Vocabulary And Study Skill

Use these words and study moves before you practise Indices and surds. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Indices and surds
- Indices
- surds
- method
- model
- unit
- answer
- check

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Indices and surds without a clear example, method, or evidence.

Indices and surds: Guided Activity

Guided activity for Indices and surds:

1. Work with a partner and read the worked example.
2. Make a similar question using Rwanda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Indices and surds: Practice And Correction

Practice for Indices and surds:

Main outcome: to be able to determine the probability of an event happening using equally likely events or experiments.

1. Write one certain event, one likely event, and one impossible event.
2. Toss a coin 10 times and record heads/tails.
3. Use your results to say what happened more often.

Answer check:

Answer check: certain means must happen; impossible means cannot happen.

Correction check:

- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Rwanda home, school, shop, garden, road, game, or timetable example connected to indices and surds and solve it.

Indices and surds: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Indices and surds.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- method is clear
- working is shown step by step
- answer is checked in context

Indices and surds: Outcome Check

Outcome check for Indices and surds: To be able to determine the probability of an event happening using equally likely events or experiments.

1. Write one clear example from the topic.
2. Solve it with labelled steps.
3. Check the answer in context.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Indices and surds: Outcome Check

Outcome check for Indices and surds: Recognize laws of indices.

1. Write one clear example from the topic.
2. Solve it with labelled steps.
3. Check the answer in context.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Polynomials: Lesson Opener

Learning focus: Polynomials

Country link: a Rwandan school, market, farming, building, transport, or data problem for Polynomials.

Progression focus: connect ideas across topics and explain causes, methods, or consequences; show methods, units, checks, and a final explanation.

By the end of this topic, you should be able to:

- to be able to perform operations, factorise polynomials, and solve related problems
- classify polynomials by degree and number of terms
- recognize operation properties on polynomials
- give common factor of algebraic expressions
- perform operation of polynomials

Inquiry questions:

- How can polynomials help you solve a real problem in Mathematics?

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Polynomials: Learn

Polynomials teaches how two quantities change together at the same rate.

Core idea:

- In direct proportion, when one quantity is multiplied, the other quantity is multiplied by the same factor.
- A table helps you compare matching values.
- The unit rate tells the value for one item, one litre, one metre, one hour, or one learner.
- Always check that the relationship is consistent before using proportion.

Key words:

- Polynomials
- method
- model
- unit
- answer
- check

Polynomials: Worked Example

Worked example for Polynomials: If 3 exercise books cost 900 francs, how much do 5 exercise books cost at the same price?

1. Find the cost of 1 book: $900 / 3 = 300$ francs.
2. Multiply by 5 books: $300 \times 5 = 1500$ francs.
3. Check: more books should cost more money.

Answer: 5 exercise books cost 1500 francs.

Polynomials: Vocabulary And Study Skill

Use these words and study moves before you practise Polynomials. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Polynomials
- method
- model
- unit
- answer
- check

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Polynomials without a clear example, method, or evidence.

Polynomials: Guided Activity

Guided activity for Polynomials:

1. Work with a partner and read the worked example.
2. Make a similar question using Rwanda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Polynomials: Practice And Correction

Practice for Polynomials:

Main outcome: to be able to perform operations, factorise polynomials, and solve related problems.

1. Complete the table: 1 pen = 200 francs, 2 pens = __, 5 pens = __.
2. A cyclist travels 18 km in 2 hours at the same speed. How far in 5 hours?
3. Write the unit rate used in each answer.

Answer check:

Answers: 2 pens = 400 francs; 5 pens = 1000 francs. The cyclist travels 45 km in 5 hours because $18 / 2 = 9$ km per hour.

Correction check:

- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Rwanda home, school, shop, garden, road, game, or timetable example connected to polynomials and solve it.

Polynomials: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Polynomials.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- method is clear
- working is shown step by step
- answer is checked in context

Polynomials: Outcome Check

Outcome check for Polynomials: To be able to perform operations, factorise polynomials, and solve related problems.

1. A recipe uses 2 cups of flour for 6 cakes. How many cups for 18 cakes?
2. Write the unit rate or scale factor.
3. Check that both quantities changed by the same factor.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Polynomials: Outcome Check

Outcome check for Polynomials: Classify polynomials by degree and number of terms.

1. A recipe uses 2 cups of flour for 6 cakes. How many cups for 18 cakes?
2. Write the unit rate or scale factor.
3. Check that both quantities changed by the same factor.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Simultaneous linear equations and inequalities: Lesson Opener

Learning focus: Simultaneous linear equations and inequalities

Country link: a Rwandan school, market, farming, building, transport, or data problem for Simultaneous linear equations and inequalities.

Progression focus: connect ideas across topics and explain causes, methods, or consequences; show methods, units, checks, and a final explanation.

By the end of this topic, you should be able to:

- to be able solve problems related to simultaneous linear equations and inequalities and represent the solutions graphically
- define simultaneous linear equations and give examples
- show whether a given simultaneous linear equation is independent, dependent, or inconsistent
- recognize the forms of compound inequalities with one unknown and give examples
- solve simultaneous linear equations in two variables
- model and solve mathematical word problems using simultaneous equations

Inquiry questions:

- How can simultaneous linear equations and inequalities help you solve a real problem in Mathematics?

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Simultaneous linear equations and inequalities: Learn

Simultaneous linear equations and inequalities teaches how lines and angles describe shapes and directions.

Core idea:

- A line can be straight, parallel, perpendicular, horizontal, vertical, or slanting.
- An angle is a turn between two lines.
- Draw and label diagrams carefully before measuring or classifying.

Key words:

- Simultaneous linear equations and inequalities
- Simultaneous
- linear
- equations
- inequalities
- method
- model
- unit

Simultaneous linear equations and inequalities: Worked Example

Worked example for Simultaneous linear equations and inequalities: A shape has one right angle. How can you show it clearly?

1. Draw the two meeting lines.
2. Mark the square corner to show a right angle.
3. Label it 90 degrees.
4. Name a classroom example, such as a book corner.

Answer: A right angle is a quarter turn, 90 degrees.

Simultaneous liner equations and inequalities: Vocabulary And Study Skill

Use these words and study moves before you practise Simultaneous liner equations and inequalities. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Simultaneous liner equations and inequalities
- Simultaneous
- liner
- equations
- inequalities
- method
- model
- unit

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Simultaneous liner equations and inequalities without a clear example, method, or evidence.

Simultaneous liner equations and inequalities: Guided Activity

Guided activity for Simultaneous liner equations and inequalities:

1. Work with a partner and read the worked example.
2. Make a similar question using Rwanda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Simultaneous liner equations and inequalities: Practice And Correction

Practice for Simultaneous liner equations and inequalities:

Main outcome: to be able solve problems related to simultaneous linear equations and inequalities and represent the solutions graphically.

1. Draw one right angle and label it 90 degrees.
2. Name one pair of parallel lines in the classroom.
3. Use a ruler to draw a straight line segment and label its endpoints.

Answer check:

Answer check: the right angle should show a square corner; parallel lines stay the same distance apart.

Correction check:

- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Rwanda home, school, shop, garden, road, game, or timetable example connected to simultaneous linear equations and inequalities and solve it.

Simultaneous linear equations and inequalities: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Simultaneous linear equations and inequalities.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- method is clear
- working is shown step by step
- answer is checked in context

Simultaneous linear equations and inequalities: Outcome Check

Outcome check for Simultaneous linear equations and inequalities: To be able solve problems related to simultaneous linear equations and inequalities and represent the solutions graphically.

1. Draw and label one right angle.
2. Name one pair of parallel lines.
3. Explain how you know.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Simultaneous linear equations and inequalities: Outcome Check

Outcome check for Simultaneous linear equations and inequalities: Define simultaneous linear equations and give examples.

1. Draw and label one right angle.
2. Name one pair of parallel lines.
3. Explain how you know.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Multiplier for proportional change: Lesson Opener

Learning focus: Multiplier for proportional change

Country link: a Rwandan school, market, farming, building, transport, or data problem for Multiplier for proportional change.

Progression focus: connect ideas across topics and explain causes, methods, or consequences; show methods, units, checks, and a final explanation.

By the end of this topic, you should be able to:

- to be able use a multiplier for proportional change
- recognize the properties of proportions
- express ratios in their simplest form
- share quantities in a given proportion or ratio
- solve problems in real life involving multiplier proportion change
- apply multipliers for proportional change to solve given problem

Inquiry questions:

- How can multiplier for proportional change help you solve a real problem in Mathematics?

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Multiplier for proportional change: Learn

Multiplier for proportional change teaches how two quantities change together at the same rate.

Core idea:

- In direct proportion, when one quantity is multiplied, the other quantity is multiplied by the same factor.
- A table helps you compare matching values.
- The unit rate tells the value for one item, one litre, one metre, one hour, or one learner.
- Always check that the relationship is consistent before using proportion.

Key words:

- Multiplier for proportional change
- Multiplier
- proportional
- change
- method
- model
- unit
- answer

Multiplier for proportional change: Worked Example

Worked example for Multiplier for proportional change: If 3 exercise books cost 900 francs, how much do 5 exercise books cost at the same price?

1. Find the cost of 1 book: $900 / 3 = 300$ francs.
2. Multiply by 5 books: $300 \times 5 = 1500$ francs.
3. Check: more books should cost more money.

Answer: 5 exercise books cost 1500 francs.

Multiplier for proportional change: Vocabulary And Study Skill

Use these words and study moves before you practise Multiplier for proportional change. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Multiplier for proportional change
- Multiplier
- proportional
- change
- method
- model
- unit
- answer

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Multiplier for proportional change without a clear example, method, or evidence.

Multiplier for proportional change: Guided Activity

Guided activity for Multiplier for proportional change:

1. Work with a partner and read the worked example.
2. Make a similar question using Rwanda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Multiplier for proportional change: Practice And Correction

Practice for Multiplier for proportional change:

Main outcome: to be able use a multiplier for proportional change.

1. Complete the table: 1 pen = 200 francs, 2 pens = __, 5 pens = __.
2. A cyclist travels 18 km in 2 hours at the same speed. How far in 5 hours?
3. Write the unit rate used in each answer.

Answer check:

Answers: 2 pens = 400 francs; 5 pens = 1000 francs. The cyclist travels 45 km in 5 hours because $18 / 2 = 9$ km per hour.

Correction check:

- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Rwanda home, school, shop, garden, road, game, or timetable example connected to multiplier for proportional change and solve it.

Multiplier for proportional change: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Multiplier for proportional change.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- method is clear
- working is shown step by step
- answer is checked in context

Multiplier for proportional change: Outcome Check

Outcome check for Multiplier for proportional change: To be able use a multiplier for proportional change.

1. A recipe uses 2 cups of flour for 6 cakes. How many cups for 18 cakes?
2. Write the unit rate or scale factor.
3. Check that both quantities changed by the same factor.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Multiplier for proportional change: Outcome Check

Outcome check for Multiplier for proportional change: Recognize the properties of proportions.

1. A recipe uses 2 cups of flour for 6 cakes. How many cups for 18 cakes?
2. Write the unit rate or scale factor.
3. Check that both quantities changed by the same factor.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Thales theorem: Lesson Opener

Learning focus: Thales theorem

Country link: a Rwandan school, market, farming, building, transport, or data problem for Thales theorem.

Progression focus: connect ideas across topics and explain causes, methods, or consequences; show methods, units, checks, and a final explanation.

By the end of this topic, you should be able to:

- to be able useThales' theorem to solve problems related to similar shapes and determinetheir lengths and areas
- identify and name triangles or trapeziums from parallel and transversals intersecting lines
- state Thales' theorem and its corollaries
- associate extended proportions in the triangles
- apply Thales' theorem and its corollaries to solve problems on proportions of triangles, trapeziums
- discuss the converse of Thales' theorem

Inquiry questions:

- How can thales theorem help you solve a real problem in Mathematics?

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Thales theorem: Learn

Thales theorem teaches how two quantities change together at the same rate.

Core idea:

- In direct proportion, when one quantity is multiplied, the other quantity is multiplied by the same factor.
- A table helps you compare matching values.
- The unit rate tells the value for one item, one litre, one metre, one hour, or one learner.
- Always check that the relationship is consistent before using proportion.

Key words:

- Thales theorem
- Thales
- theorem
- method
- model
- unit
- answer
- check

Thales theorem: Worked Example

Worked example for Thales theorem: If 3 exercise books cost 900 francs, how much do 5 exercise books cost at the same price?

1. Find the cost of 1 book: $900 / 3 = 300$ francs.
2. Multiply by 5 books: $300 \times 5 = 1500$ francs.
3. Check: more books should cost more money.

Answer: 5 exercise books cost 1500 francs.

Thales theorem: Vocabulary And Study Skill

Use these words and study moves before you practise Thales theorem. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Thales theorem
- Thales
- theorem
- method
- model
- unit
- answer
- check

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Thales theorem without a clear example, method, or evidence.

Thales theorem: Guided Activity

Guided activity for Thales theorem:

1. Work with a partner and read the worked example.
2. Make a similar question using Rwanda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Thales theorem: Practice And Correction

Practice for Thales theorem:

Main outcome: to be able useThales' theorem to solve problems related to similar shapes and determinetheir lengths and areas.

1. Complete the table: 1 pen = 200 francs, 2 pens = __, 5 pens = __.
2. A cyclist travels 18 km in 2 hours at the same speed. How far in 5 hours?
3. Write the unit rate used in each answer.

Answer check:

Answers: 2 pens = 400 francs; 5 pens = 1000 francs. The cyclist travels 45 km in 5 hours because $18 / 2 = 9$ km per hour.

Correction check:

- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Rwanda home, school, shop, garden, road, game, or timetable example connected to Thales theorem and solve it.

Thales theorem: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Thales theorem.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- method is clear
- working is shown step by step
- answer is checked in context

Thales theorem: Outcome Check

Outcome check for Thales theorem: To be able use Thales' theorem to solve problems related to similar shapes and determine their lengths and areas.

1. A recipe uses 2 cups of flour for 6 cakes. How many cups for 18 cakes?
2. Write the unit rate or scale factor.
3. Check that both quantities changed by the same factor.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Thales theorem: Outcome Check

Outcome check for Thales theorem: Identify and name triangles or trapeziums from parallel and transversals intersecting lines.

1. A recipe uses 2 cups of flour for 6 cakes. How many cups for 18 cakes?
2. Write the unit rate or scale factor.
3. Check that both quantities changed by the same factor.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Pythagoras's theorem: Lesson Opener

Learning focus: Pythagoras's theorem

Country link: a Rwandan school, market, farming, building, transport, or data problem for Pythagoras's theorem.

Progression focus: connect ideas across topics and explain causes, methods, or consequences; show methods, units, checks, and a final explanation.

By the end of this topic, you should be able to:

- find the length of each side of a right angle triangles using Pythagoras' theorem
- state Pythagoras' theorem
- identify the hypotenuse from the three sides of a right angle triangle
- list properties of a right angled triangle
- use Pythagoras' theorem to find the lengths of the sides of right angle triangles
- apply Pythagoras' theorem to solve problems in a range of contexts

Inquiry questions:

- How can pythagoras's theorem help you solve a real problem in Mathematics?

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Pythagoras's theorem: Learn

Pythagoras's theorem teaches how to measure distance or how long something is.

Core idea:

- Choose the unit that fits the object: millimetres, centimetres, metres, or kilometres.
- Convert by knowing the relationship between units.
- Estimate before measuring so you can notice unreasonable answers.

Key words:

- Pythagoras's theorem
- Pythagoras's
- theorem
- method
- model
- unit
- answer
- check

Pythagoras's theorem: Worked Example

Worked example for Pythagoras's theorem: A classroom is 8 m long and a corridor is 350 cm long. Which is longer?

1. Convert 8 m to centimetres: $8\text{ m} = 800\text{ cm}$.
2. Compare 800 cm and 350 cm.
3. 800 cm is greater than 350 cm.

Answer: The classroom is longer.

Pythagoras's theorem: Vocabulary And Study Skill

Use these words and study moves before you practise Pythagoras's theorem. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Pythagoras's theorem
- Pythagoras's
- theorem
- method
- model
- unit
- answer
- check

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Pythagoras's theorem without a clear example, method, or evidence.

Pythagoras's theorem: Guided Activity

Guided activity for Pythagoras's theorem:

1. Work with a partner and read the worked example.
2. Make a similar question using Rwanda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Pythagoras's theorem: Practice And Correction

Practice for Pythagoras's theorem:

Main outcome: find the length of each side of a right angle triangles using Pythagoras' theorem.

1. Estimate the length of a desk, then choose a suitable unit.
2. Convert 4 m to centimetres.
3. Solve: a path is 120 m and another is 75 m. What is the total length?

Answer check:

Answer: 4 m = 400 cm. $120\text{ m} + 75\text{ m} = 195\text{ m}$.

Correction check:

- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Rwanda home, school, shop, garden, road, game, or timetable example connected to pythagoras's theorem and solve it.

Pythagoras's theorem: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Pythagoras's theorem.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- method is clear
- working is shown step by step
- answer is checked in context

Pythagoras's theorem: Outcome Check

Outcome check for Pythagoras's theorem: Find the length of each side of a right angle triangles using Pythagoras' theorem.

1. Convert 6 m to centimetres.
2. Find the total length of 45 m and 38 m.
3. State the unit clearly.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Pythagoras's theorem: Outcome Check

Outcome check for Pythagoras's theorem: State Pythagoras' theorem.

1. Convert 6 m to centimetres.
2. Find the total length of 45 m and 38 m.
3. State the unit clearly.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Vectors: Lesson Opener

Learning focus: Vectors

Country link: a Rwandan school, market, farming, building, transport, or data problem for Vectors.

Progression focus: connect ideas across topics and explain causes, methods, or consequences; show methods, units, checks, and a final explanation.

By the end of this topic, you should be able to:

- solve problems doing operation on vectors
- represent a vector in a Cartesian plane
- differentiate between vector quantities and scalar quantities
- show whether vectors are equal
- use vector notations correctly and perform operations on vectors
- solve and explain problems about vectors

Inquiry questions:

- How can vectors help you solve a real problem in Mathematics?

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Vectors: Learn

Vectors teaches how two quantities change together at the same rate.

Core idea:

- In direct proportion, when one quantity is multiplied, the other quantity is multiplied by the same factor.
- A table helps you compare matching values.
- The unit rate tells the value for one item, one litre, one metre, one hour, or one learner.
- Always check that the relationship is consistent before using proportion.

Key words:

- Vectors
- method
- model
- unit
- answer
- check

Vectors: Worked Example

Worked example for Vectors: If 3 exercise books cost 900 francs, how much do 5 exercise books cost at the same price?

1. Find the cost of 1 book: $900 \div 3 = 300$ francs.
2. Multiply by 5 books: $300 \times 5 = 1500$ francs.
3. Check: more books should cost more money.

Answer: 5 exercise books cost 1500 francs.

Vectors: Vocabulary And Study Skill

Use these words and study moves before you practise Vectors. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Vectors
- method
- model
- unit
- answer
- check

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Vectors without a clear example, method, or evidence.

Vectors: Guided Activity

Guided activity for Vectors:

1. Work with a partner and read the worked example.
2. Make a similar question using Rwanda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Vectors: Practice And Correction

Practice for Vectors:

Main outcome: solve problems doing operation on vectors.

1. Complete the table: 1 pen = 200 francs, 2 pens = __, 5 pens = __.
2. A cyclist travels 18 km in 2 hours at the same speed. How far in 5 hours?
3. Write the unit rate used in each answer.

Answer check:

Answers: 2 pens = 400 francs; 5 pens = 1000 francs. The cyclist travels 45 km in 5 hours because $18 / 2 = 9$ km per hour.

Correction check:

- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Rwanda home, school, shop, garden, road, game, or timetable example connected to vectors and solve it.

Vectors: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Vectors.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- method is clear
- working is shown step by step
- answer is checked in context

Vectors: Outcome Check

Outcome check for Vectors: Solve problems doing operation on vectors.

1. A recipe uses 2 cups of flour for 6 cakes. How many cups for 18 cakes?
2. Write the unit rate or scale factor.
3. Check that both quantities changed by the same factor.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Vectors: Outcome Check

Outcome check for Vectors: Represent a vector in a Cartesian plane.

1. A recipe uses 2 cups of flour for 6 cakes. How many cups for 18 cakes?
2. Write the unit rate or scale factor.
3. Check that both quantities changed by the same factor.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Parallel and orthogonal projections: Lesson Opener

Learning focus: Parallel and orthogonal projections

Country link: a Rwandan school, market, farming, building, transport, or data problem for Parallel and orthogonal projections.

Progression focus: connect ideas across topics and explain causes, methods, or consequences; show methods, units, checks, and a final explanation.

By the end of this topic, you should be able to:

- transform shapes under parallel or orthogonal projections
- identify an image of a figure under parallel projection
- identify an image of a figure under orthogonal projection
- construct an image of an object or geometric shape under parallel projection and orthogonal projection
- show the importance of parallel and orthogonal projection in various situations
- develop critical thinking and reasoning skills, while transforming shapes under parallel or orthogonal projection

Inquiry questions:

- How can parallel and orthogonal projections help you solve a real problem in Mathematics?

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Parallel and orthogonal projections: Learn

Parallel and orthogonal projections teaches properties of two-dimensional shapes.

Core idea:

- Shapes can be classified by sides, corners, lines of symmetry, and angles.
- A labelled sketch helps you compare shapes accurately.
- Use correct names such as triangle, rectangle, square, and circle.

Key words:

- Parallel and orthogonal projections
- Parallel
- orthogonal
- projections
- method
- model
- unit
- answer

Parallel and orthogonal projections: Worked Example

Worked example for Parallel and orthogonal projections: Classify a shape with 4 equal sides and 4 right angles.

1. Four sides means it is a quadrilateral.
2. Equal sides and right angles match a square.
3. Check the name and properties.

Answer: The shape is a square.

Parallel and orthogonal projections: Vocabulary And Study Skill

Use these words and study moves before you practise Parallel and orthogonal projections. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Parallel and orthogonal projections
- Parallel
- orthogonal
- projections
- method
- model
- unit
- answer

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Parallel and orthogonal projections without a clear example, method, or evidence.

Parallel and orthogonal projections: Guided Activity

Guided activity for Parallel and orthogonal projections:

1. Work with a partner and read the worked example.
2. Make a similar question using Rwanda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Parallel and orthogonal projections: Practice And Correction

Practice for Parallel and orthogonal projections:

Main outcome: transform shapes under parallel or orthogonal projections.

1. Draw a triangle, rectangle, and square.
2. Write two properties for each shape.
3. Circle the shape that has four equal sides and four right angles.

Answer check:

Answer: the square has four equal sides and four right angles.

Correction check:

- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Rwanda home, school, shop, garden, road, game, or timetable example connected to parallel and orthogonal projections and solve it.

Parallel and orthogonal projections: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Parallel and orthogonal projections.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- method is clear
- working is shown step by step
- answer is checked in context

Parallel and orthogonal projections: Outcome Check

Outcome check for Parallel and orthogonal projections: Transform shapes under parallel or orthogonal projections.

1. Draw a square and a rectangle.
2. Write two properties they share.
3. Write one property that is different.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Parallel and orthogonal projections: Outcome Check

Outcome check for Parallel and orthogonal projections: Identify an image of a figure under parallel projection.

1. Draw a square and a rectangle.
2. Write two properties they share.
3. Write one property that is different.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Isometries: Lesson Opener

Learning focus: Isometries

Country link: a Rwandan school, market, farming, building, transport, or data problem for Isometries.

Progression focus: connect ideas across topics and explain causes, methods, or consequences; show methods, units, checks, and a final explanation.

By the end of this topic, you should be able to:

- transform shapes using congruence, central symmetry, reflection, translation, and rotation
- identify an image of a figure under central symmetry, reflection, translation, rotation
- construct the image of a point, a segment, a geometric shape, under central symmetry, reflection, translation, rotation
- find the coordinates of an image of an object under central symmetry, reflection, translation, rotation
- appreciate that translation, rotation and reflection play important roles in various situations
- develop team spirit

Inquiry questions:

- How can isometries help you solve a real problem in Mathematics?

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Isometries: Learn

Isometries teaches properties of two-dimensional shapes.

Core idea:

- Shapes can be classified by sides, corners, lines of symmetry, and angles.
- A labelled sketch helps you compare shapes accurately.
- Use correct names such as triangle, rectangle, square, and circle.

Key words:

- Isometries
- method
- model
- unit

Isometries: Worked Example

Worked example for Isometries: Classify a shape with 4 equal sides and 4 right angles.

1. Four sides means it is a quadrilateral.
2. Equal sides and right angles match a square.
3. Check the name and properties.

Answer: The shape is a square.

Isometries: Vocabulary And Study Skill

Use these words and study moves before you practise Isometries. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Isometries
- method
- model
- unit
- answer
- check

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Isometries without a clear example, method, or evidence.

Isometries: Guided Activity

Guided activity for Isometries:

1. Work with a partner and read the worked example.
2. Make a similar question using Rwanda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Isometries: Practice And Correction

Practice for Isometries:

Main outcome: transform shapes using congruence, central symmetry, reflection, translation, and rotation.

1. Draw a triangle, rectangle, and square.
2. Write two properties for each shape.
3. Circle the shape that has four equal sides and four right angles.

Answer check:

Answer: the square has four equal sides and four right angles.

Correction check:

- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Rwanda home, school, shop, garden, road, game, or timetable example connected to isometries and solve it.

Isometries: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Isometries.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- method is clear
- working is shown step by step
- answer is checked in context

Isometries: Outcome Check

Outcome check for Isometries: Transform shapes using congruence, central symmetry, reflection, translation, and rotation.

1. Draw a square and a rectangle.
2. Write two properties they share.
3. Write one property that is different.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Isometries: Outcome Check

Outcome check for Isometries: Identify an image of a figure under central symmetry, reflection, translation, rotation.

1. Draw a square and a rectangle.
2. Write two properties they share.
3. Write one property that is different.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Statistics (grouped data): Lesson Opener

Learning focus: Statistics (grouped data)

Country link: a Rwandan school, market, farming, building, transport, or data problem for Statistics (grouped data).

Progression focus: connect ideas across topics and explain causes, methods, or consequences; show methods, units, checks, and a final explanation.

By the end of this topic, you should be able to:

- collect, represent, and interpret grouped data
- define grouped data and represent grouped data on a frequency distribution
- identify mode, middle class, modal class, and median of given grouped statistical data
- read diagram of grouped statistical data
- apply data collection to carry out research
- represent grouped statistical information using histograms, polygons, frequency distribution tables, and pie charts

Inquiry questions:

- How can statistics (grouped data) help you solve a real problem in Mathematics?

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Statistics (grouped data): Learn

Statistics (grouped data) teaches how to collect, organize, read, and explain information.

Core idea:

- Data must answer a clear question.
- Tables, tally charts, pictographs, and bar charts make data easier to read.
- A good conclusion says what the data shows.

Key words:

- Statistics (grouped data)
- Statistics
- grouped
- data
- method
- model
- unit
- answer

Statistics (grouped data): Worked Example

Worked example for Statistics (grouped data): A class records favourite fruits: mango 8, banana 12, pineapple 6. Which fruit is most popular?

1. Read the table values.
2. Compare 8, 12, and 6.
3. 12 is the greatest number.

Answer: Banana is most popular.

Statistics (grouped data): Vocabulary And Study Skill

Use these words and study moves before you practise Statistics (grouped data). Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Statistics (grouped data)
- Statistics
- grouped
- data
- method
- model
- unit
- answer

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Statistics (grouped data) without a clear example, method, or evidence.

Statistics (grouped data): Guided Activity

Guided activity for Statistics (grouped data):

1. Work with a partner and read the worked example.
2. Make a similar question using Rwanda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Statistics (grouped data): Practice And Correction

Practice for Statistics (grouped data):

Main outcome: collect, represent, and interpret grouped data.

1. Ask 10 learners a simple question and make a tally table.
2. Draw a bar chart from the tally table.
3. Write one conclusion from the data.

Answer check:

Answer check: totals in the tally table and bar chart should match.

Correction check:

- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Rwanda home, school, shop, garden, road, game, or timetable example connected to statistics (grouped data) and solve it.

Statistics (grouped data): Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Statistics (grouped data).
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- method is clear
- working is shown step by step
- answer is checked in context

Statistics (grouped data): Outcome Check

Outcome check for Statistics (grouped data): Collect, represent, and interpret grouped data.

1. Write one clear example from the topic.
2. Solve it with labelled steps.
3. Check the answer in context.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Statistics (grouped data): Outcome Check

Outcome check for Statistics (grouped data): Define grouped data and represent grouped data on a frequency distribution.

1. Write one clear example from the topic.
2. Solve it with labelled steps.
3. Check the answer in context.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Tree and Venn diagrams: Lesson Opener

Learning focus: Tree and Venn diagrams

Country link: a Rwandan school, market, farming, building, transport, or data problem for Tree and Venn diagrams.

Progression focus: connect ideas across topics and explain causes, methods, or consequences; show methods, units, checks, and a final explanation.

By the end of this topic, you should be able to:

- determine probabilities and assess likelihood by using tree and venn diagrams
- define mutually exclusive and independent events
- count the number of branches and total number of outcomes on a tree diagram
- construct and interpret a tree diagram correctly
- use Tree and Venn diagrams to determine probability
- appreciate the importance of probability to find the chance happening for an event to happen

Inquiry questions:

- How can tree and venn diagrams help you solve a real problem in Mathematics?

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Tree and Venn diagrams: Learn

Tree and Venn diagrams teaches how heavy objects are.

Core idea:

- Mass is measured with units such as grams and kilograms.
- A balance or scale helps compare mass fairly.
- Convert units before adding or comparing if the units are different.

Key words:

- Tree and Venn diagrams
- Tree
- Venn
- diagrams
- method
- model
- unit
- answer

Tree and Venn diagrams: Worked Example

Worked example for Tree and Venn diagrams: A bag of beans has a mass of 3 kg. A small packet has a mass of 500 g. What is the total mass in grams?

1. Convert 3 kg to grams: $3 \text{ kg} = 3000 \text{ g}$.
2. Add $3000 \text{ g} + 500 \text{ g} = 3500 \text{ g}$.

Answer: 3500 g.

Tree and Venn diagrams: Vocabulary And Study Skill

Use these words and study moves before you practise Tree and Venn diagrams. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Tree and Venn diagrams
- Tree
- Venn
- diagrams
- method
- model
- unit
- answer

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Tree and Venn diagrams without a clear example, method, or evidence.

Tree and Venn diagrams: Guided Activity

Guided activity for Tree and Venn diagrams:

1. Work with a partner and read the worked example.
2. Make a similar question using Rwanda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Tree and Venn diagrams: Practice And Correction

Practice for Tree and Venn diagrams:

Main outcome: determine probabilities and assess likelihood by using tree and venn diagrams.

1. Name two objects measured in grams and two measured in kilograms.
2. Convert 2 kg to grams.
3. Compare 1500 g and 2 kg. Which is heavier?

Answer check:

Answer: $2\text{ kg} = 2000\text{ g}$, so 2 kg is heavier than 1500 g.

Correction check:

- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Rwanda home, school, shop, garden, road, game, or timetable example connected to tree and venn diagrams and solve it.

Tree and Venn diagrams: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Tree and Venn diagrams.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- method is clear
- working is shown step by step
- answer is checked in context

Tree and Venn diagrams: Outcome Check

Outcome check for Tree and Venn diagrams: Determine probabilities and assess likelihood by using tree and venn diagrams.

1. Convert 4 kg to grams.
2. Compare 2500 g and 3 kg.
3. Write the heavier mass with a reason.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Tree and Venn diagrams: Outcome Check

Outcome check for Tree and Venn diagrams: Define mutually exclusive and independent events.

1. Convert 4 kg to grams.
2. Compare 2500 g and 3 kg.
3. Write the heavier mass with a reason.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Indices and surds: Review Clinic 1

Use this review clinic to strengthen Mathematics without copying earlier answers.

1. Choose one idea from Indices and surds that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Rwanda.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

Polynomials: Review Clinic 2

Use this review clinic to strengthen Mathematics without copying earlier answers.

1. Choose one idea from Polynomials that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Rwanda.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

Simultaneous liner equations and inequalities: Review Clinic 3

Use this review clinic to strengthen Mathematics without copying earlier answers.

1. Choose one idea from Simultaneous liner equations and inequalities that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Rwanda.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

Multiplier for proportional change: Review Clinic 4

Use this review clinic to strengthen Mathematics without copying earlier answers.

1. Choose one idea from Multiplier for proportional change that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Rwanda.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

Glossary

Use this glossary to revise important words in Mathematics.

- Indices and surds: Explain this term using an example from the book, your school, or your community.
- Polynomials: Explain this term using an example from the book, your school, or your community.
- Simultaneous linear equations and inequalities: Explain this term using an example from the book, your school, or your community.
- Multiplier for proportional change: Explain this term using an example from the book, your school, or your community.
- Thales theorem: Explain this term using an example from the book, your school, or your community.
- Pythagoras's theorem: Explain this term using an example from the book, your school, or your community.
- Vectors: Explain this term using an example from the book, your school, or your community.
- Parallel and orthogonal projections: Explain this term using an example from the book, your school, or your community.
- Isometries: Explain this term using an example from the book, your school, or your community.
- Statistics (grouped data): Explain this term using an example from the book, your school, or your community.
- Tree and Venn diagrams: Explain this term using an example from the book, your school, or your community.

Add five more words from your lessons and write your own meanings.

Answer And Teacher Notes

A strong Mathematics answer shows the model, method, units or labels, final answer, and a check. Use the worked examples and answer checks in each practice page to correct your work. When an answer is wrong, find the first step that changed the meaning and correct that step before trying a new question.

Final Project

Create a final project for Mathematics that shows what you can now do independently.

Your project must include:

1. A clear title.
2. A real-life problem, question, text, performance, investigation, design, or worked task.
3. Evidence from at least three lessons in this book.
4. A drawing, table, map, chart, model, dialogue, paragraph, performance plan, practical product, or worked solution.
5. A correction note showing one improvement after feedback.
6. A short reflection explaining what you learned, what was difficult, and what you will practise next.

Presentation check: keep the work neat, label important parts, use correct vocabulary, and be ready to explain your choices to a classmate or teacher.